Question 1

Trace the network path to a website and count how many intermediate routers are involved.

Approach

Use network diagnostic tool to view packet route and identify hop count.

Commands

- `traceroute` → shows network path
- `traceroute domain` → displays hop sequence

Output

STEP 1 : run traceroute

```
┌──(aman㉿kali)-[~]
└─$ traceroute google.com
traceroute to google.com (172.217.167.238), 30 hops max, 60 byte packets
 1  10.0.2.2 (10.0.2.2)  0.319 ms  0.516 ms  0.483 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  * * *
 8  * * *
 9  * * *
10  * * *
11  * * *
12  * * *
13  * * *
14  * * *
15  * * *
16  * * *
17  * * *
18  * * *
19  * * *
20  * * *
21  * * *
22  * * *
23  * * *
24  * * *
25  * * *
26  * * *
27  * * *
28  * * *
29  * * *
30  * * *
```

Key Observations

Each hop represents a router between source and destination.
Higher hops may indicate longer routing path.

What I learned

Network paths reveal infrastructure between user and website.

Question 2

Install Tor browser and compare browsing speed with normal browsing.

Approach

Install Tor package and open browser to test anonymous browsing performance.

Commands

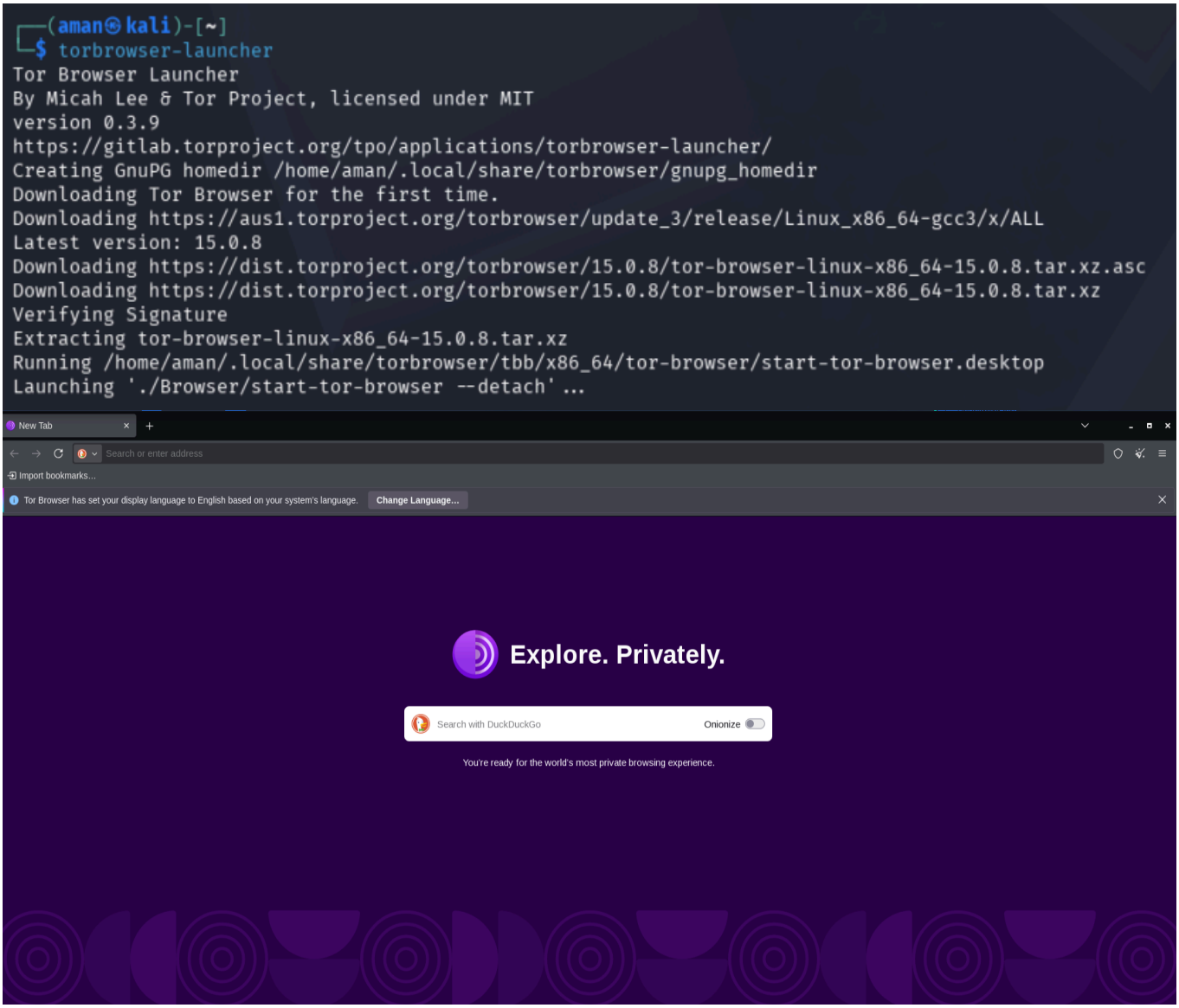
- `apt install` → install software
- `torbrowser-launcher` → launch tor browser

Output

STEP 1 : install tor browser

```
┌──(aman㉿kali)-[~]
└─$ sudo apt install torbrowser-launcher
[sudo] password for aman:
The following packages were automatically installed and are no longer required:
  curlftpfs      libavformat61              libfuse2t64            libmpeg2encpp-2.1-0t64  libplacebo351     librubberband2  libsphinxbase3t64  mesa-vdpau-drivers  python3-aiomcache       python3-wapiti-arsenic  vdpau-driver-all
  libaudio2      libconfig-inifiles-perl    libgav1-1              libmplex2-2.1-0t64      libpocketsphinx3  libsframe2      libswscale8        pocketsphinx-en-us  python3-fs              python3-yaswfp
  libavfilter10  libcrypt-dev               libmjpegutils-2.1-0t64 libmupdf25.1            libpostproc58     libsnmp40t64    libvdpau-va-gl1    python3-aiocache    python3-terminaltables  ruby-unf-ext
Use 'sudo apt autoremove' to remove them.

Installing:
  torbrowser-launcher

Installing dependencies:
  libtorsocks  python3-gpg  python3-pyside6.qtcore  python3-pyside6.qtgui  python3-pyside6.qtwidgets  tor  tor-geoipdb  torsocks

Suggested packages:
  python-pyside6-doc  mixmaster  apparmor-utils  nyx  obfs4proxy

Summary:
  Upgrading: 0, Installing: 9, Removing: 0, Not Upgrading: 565
  Download size: 8,597 kB
  Space needed: 51.8 MB / 11.1 GB available

Continue? [Y/n] y
Get:1 http://kali.download/kali kali-rolling/main amd64 libtorsocks amd64 2.5.0-6 [46.0 kB]
Get:2 http://http.kali.org/kali kali-rolling/main amd64 python3-gpg amd64 2.0.0-2+b1 [277 kB]
Get:3 http://http.kali.org/kali kali-rolling/main amd64 python3-pyside6.qtcore amd64 6.9.2-2+b1 [959 kB]
Get:7 http://mirrors.esto.network/kali kali-rolling/main amd64 tor-geoipdb all 0.4.9.5-2 [2,685 kB]
Get:4 http://http.kali.org/kali kali-rolling/main amd64 python3-pyside6.qtgui amd64 6.9.2-2+b1 [1,164 kB]
Get:5 http://http.kali.org/kali kali-rolling/main amd64 python3-pyside6.qtwidgets amd64 6.9.2-2+b1 [1,296 kB]
Get:6 http://kali.download/kali kali-rolling/main amd64 tor amd64 0.4.9.5-2 [2,083 kB]
Get:8 http://kali.download/kali kali-rolling/contrib amd64 torbrowser-launcher amd64 0.3.9-1 [57.1 kB]
Get:9 http://kali.download/kali kali-rolling/main amd64 torsocks all 2.5.0-6 [27.9 kB]
Fetched 8,597 kB in 3s (3,310 kB/s)
Selecting previously unselected package libtorsocks:amd64.
(Reading database… 438753 files and directories currently installed.)
Preparing to unpack …/0-libtorsocks_2.5.0-6_amd64.deb…
Unpacking libtorsocks:amd64 (2.5.0-6)…
Selecting previously unselected package python3-gpg.
Preparing to unpack …/1-python3-gpg_2.0.0-2+b1_amd64.deb…
Unpacking python3-gpg (2.0.0-2+b1)…
Selecting previously unselected package python3-pyside6.qtcore.
Preparing to unpack …/2-python3-pyside6.qtcore_6.9.2-2+b1_amd64.deb…
Unpacking python3-pyside6.qtcore (6.9.2-2+b1)…
Selecting previously unselected package python3-pyside6.qtgui.
Preparing to unpack …/3-python3-pyside6.qtgui_6.9.2-2+b1_amd64.deb…
Unpacking python3-pyside6.qtgui (6.9.2-2+b1)…
Selecting previously unselected package python3-pyside6.qtwidgets.
Preparing to unpack …/4-python3-pyside6.qtwidgets_6.9.2-2+b1_amd64.deb…
Unpacking python3-pyside6.qtwidgets (6.9.2-2+b1)…
Selecting previously unselected package tor.
```

STEP 2 : launch tor browser



Key Observations

Tor routes traffic through multiple encrypted relays.
Speed is slower due to multiple hops.

What I learned

Anonymity networks trade speed for privacy.

Question 3

Configure proxychains with Firefox and browse a website through proxy routing.

Approach

Use proxychains tool to route browser traffic through chained proxies.

Commands

- `proxychains` → route traffic through proxy
- `firefox` → launch browser
- `nano` → edit proxy configuration

Output

STEP 1 : edit proxychains config

```
┌──(aman㉿kali)-[~]
└─$ sudo nano /etc/proxychains.conf
```

STEP 2 : launch firefox using proxychains

```
┌──(aman㉿kali)-[~]
└─$ proxychains firefox
[proxychains] config file found: /etc/proxychains.conf
[proxychains] preloading /usr/lib/x86_64-linux-gnu/libproxychains.so.4
[proxychains] DLL init: proxychains-ng 4.17
[proxychains] DLL init: proxychains-ng 4.17
[proxychains] DLL init: proxychains-ng 4.17
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[proxychains] DLL init: proxychains-ng 4.17
[proxychains] Strict chain  ...  127.0.0.1:9050  ...  timeout
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[proxychains] Strict chain  ...  127.0.0.1:9050  ...  timeout
```

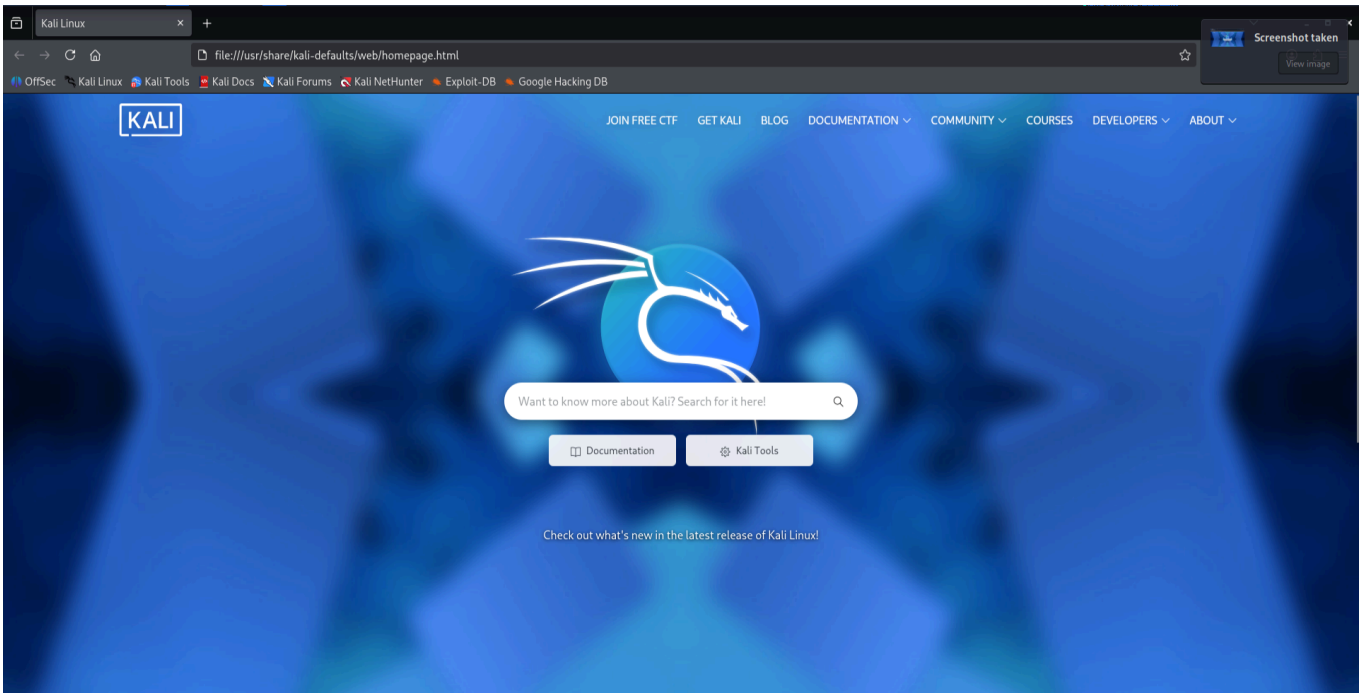

Key Observations

Proxychains hides original IP address.
Multiple proxies increase anonymity.

What I learned

Traffic obfuscation helps reduce traceability.

Security Insights

1. Traceroute exposes network structure useful for reconnaissance.
2. Tor hides identity by routing traffic through anonymous nodes.
3. Proxychains masks IP by chaining multiple proxies.